Given the above two types of prompt transformations, we can define narrower types of LLM self-consistency.

Definition 2.5 (Hypothetical consistency)**.** A model p_θ is *hypothetically consistent* if $g_{p_\theta}(p) \sim g_{p_\theta}(f(p', p))$ for any prompt $p \in \mathcal{V}^*$ and hypothetical transformation prompt template $p' \in \mathcal{P}'_I$.

Claim 2.6. If p_θ is self-consistent, then p_θ is also hypothetically consistent.

Proof. Consider prompt $p \in \mathcal{V}^*$ and hypothetical transformation prompt template $p \in \mathcal{P}'_I$. Since $f(p', p) \sim f(_, p)$ (by Definition 2.2) and $f(_, p) \sim p$, then $f(p', p) \sim p$ by the transitive property of $\sim$. Then by Definition 2.1, it follows that $g_{p_\theta}(f(p', p)) \sim g_{p_\theta}(p)$. □

Definition 2.7 (Consistency over compositional transformations)**.** A model p_θ is *compositionally consistent* when, for all pairs of compositional prompts $p \in \mathcal{P}_{\text{Comp}}$ and $f(p', p) \in \mathcal{P}_{\text{Comp}}$,

1. $p \sim g_{p_\theta}(p)$ and $g_{p_\theta}(f(p', p)) \sim g_{p_\theta}(f(p', g_{p_\theta}(p)))$

and is *compositionally inconsistent* when:

1. $p \nsim g_{p_\theta}(p)$ and $g_{p_\theta}(f(p', p)) \sim g_{p_\theta}(f(p', g_{p_\theta}(p)))$
2. $p \sim g_{p_\theta}(p)$ and $g_{p_\theta}(f(p', p)) \nsim g_{p_\theta}(f(p', g_{p_\theta}(p)))$

One more case exists that we do not count into our measures of either compositional consistency or inconsistency. If $p \nsim g_{p_\theta}(p)$ and $g_{p_\theta}(f(p', p)) \nsim g_{p_\theta}(f(p', g_{p_\theta}(p)))$, then we cannot say this is compositionally consistent because the model's output for p does not necessarily relate to or imply its output to $f(p', p)$. However, we also cannot necessarily say that it is compositionally inconsistent to give nonequivalent responses to $f(p', p)$ and $f(p', g_{p_\theta}(p))$ if the model's responses to p and $g_{p_\theta}(p)$ are also nonequivalent.

3 Evaluating Consistency on Hypothetical Transformations

In our above definitions, hypothetical consistency is characterized as a binary – either a model is hypothetically consistent across *all* pairs $(p, p') \in \mathcal{V}^* \times \mathcal{P}'_I$ or it is hypothetically inconsistent. But in practice, models are only hypothetically consistent in some cases, and it is likely impossible to achieve hypothetical consistency across all $(p, p') \in \mathcal{V}^* \times \mathcal{P}'_I$. Instead, we explore the *degree* to which LLM outputs are invariant to hypothetical transformations of the prompt. This is measured as the **hypothetical consistency rate**, which is the proportion of pairs $(p, p') \in \mathcal{V}^* \times \mathcal{P}'_I$ for which a model p_θ exhibits the property $g_{p_\theta}(p) \sim g_{p_\theta}(f(p', p))$. To measure hypothetical consistency rate, we devise a set of four hypothetical transformation prompt templates that we use to transform randomly sampled prompts (sourced from Wikipedia and DailyDialog) into hypothetical prompts, as shown in Table 1. We average the hypothetical consistency rate over these prompts to mitigate the model's sensitivity to prompt wording (Elazar et al., 2021).

Evaluating hypothetical consistency requires checking whether $g_{p_\theta}(p) \sim g_{p_\theta}(f(p', p))$. However, checking this semantic equivalence is non-trivial – the same idea can be expressed in a number of different but synonymous ways. Rather than attempt to devise an automatic method for evaluating semantic equivalence, we instead use a multiple-choice set-up. One answer choice is the continuation of the initial prompt (denoted by "`<prompt>`") sourced from a text dataset, one choice is the model's own greedily decoded completion for `<prompt>`, and the three remaining choices are the other models' completions for `<prompt>`. As discussed before, a model that is hypothetically consistent can, in a sense, predict its own completion. Thus, the model should be more likely to generate the answer choice that corresponds to its own completion than to the other answer choices. These templates are designed both to query the model on what its completion would hypothetically be for a given prompt and to evaluate whether the model can distinguish its own completions from those of other models.

As an example, suppose the original prompt sourced from Wikipedia is "This quilt begun in 1856 when she was seventeen includes the autographs on top of the blocks of many known celebrities and politicians of the day. Other". Suppose that the first three words of the completions generated by OpenAI GPT-3 models `ada-001`, `babbage-001`, `curie-001`, and `davinci-003` are "notable quilt authors," "famous quilts include,"

"signatures include abolitionists," and "notable figures whose," respectively. Additionally, the next three words of the Wikipedia article are "figures represented on." Then a hypothetical transformation prompt that uses the first template in Table 1 might look like:

> I predict that the next 3 words after "This quilt begun in 1856 when she was seventeen includes the autographs on top of the blocks of many known celebrities and politicians of the day. Other" would be
>
> A) famous quilts include
> B) figures represented on
> C) notable quilt authors
> D) signatures include abolitionists
> E) notable figures whose
>
> Answer:

If the model being evaluated is `davinci-003`, then the correct answer would be E. In this context, $g_{\texttt{davinci-003}}$("I predict that... Other" would be") = "E" $\sim$ "notable figures whose" = $g_{\texttt{davinci-003}}$("This quilt begun... Other"), which satisfies Definition 2.5. As mentioned in Section 3.1, all hypothetical prompts are few-shot (where the provided labels are the answer choices that correspond to the evaluated model's completion), and only case-insensitive exact-match answers (*i.e.* "a/A/b/B/c/C/d/D") are accepted as correct.

Although this experimental set-up eases the burden of checking the semantic equivalence of $g_{p_\theta}(p) \sim g_{p_\theta}(f(p', p))$, it is likely still a lower bound on the true hypothetical consistency, since it does not account for other model outputs that may be semantically equivalent to $g_{p_\theta}(p)$. In the example above, had `davinci-003` generated "notable people whose" instead of the letter "E," this response would have still been marked incorrect despite the semantic equivalence of "notable figures whose" and "notable people whose." Furthermore, the definition of semantic equivalence here depends implicitly on the prompt p. Nevertheless, we still find it useful to analyze these lower bounds – examinations of the failure modes of LLMs can aid in the future improvements of these models. We leave to future work the question of how far this lower bound is from the true hypothetical consistency rate.

Table 1: Prompt templates used to evaluate whether LLMs are consistent across hypothetical prompt transformations. `<prompt>` is a prompt sourced from a dataset (e.g. Wikipedia, DailyDialog), `<m>` is the number of words of its own completion that the model is asked to predict, and `<answer_choices>` are the multiple-choice answer choices that the model is given.

Hypothetical Transformation Prompt Templates
"I predict that the next `<m>` words after "`<prompt>`" would be `<answer_choices>`. Answer:"
"Given the prompt "`<prompt>`", my next `<m>` words would be `<answer_choices>`. Answer:"
"Given the context "`<prompt>`", my next `<m>` words would be `<answer_choices>`. Answer:"
"I predict that after the context "`<prompt>`" the next `<m>` words I would say are `<answer_choices>`. Answer:"

We conduct our hypothetical consistency experiments with original prompts sourced from two language modeling tasks:

- **Wikipedia** Since language models are frequently pre-trained on Wikipedia archives, evaluating on a Wikipedia dataset can confound information memorized during pre-training with the skill being evaluated. To address this issue, we collect a sample of 400 English Wikipedia (Wikimedia Foundation) articles that were created on or after June 30, 2021, since the OpenAI documentation (Center) indicates that the latest pre-training data for the `ada-`/`babbage-`/`curie-`/`davinci-001` and `davinci-003` models contains documents dating up to June 2021. Each initial prompt is a randomly

selected segment of a Wikipedia article consisting of two full sentences followed by a random-length prefix of the next sentence.

- **DailyDialog**: DailyDialog (Li et al., 2017) is a manually labeled dataset of multi-turn conversations about daily life. We choose this dataset because it contains language that is more colloquial in style and less factual in content than the Wikipedia dataset. We randomly sample 400 examples from the training split and use the first conversational turn as the initial prompt.

We use only the prompts for which all five answer choices are distinct. We also vary the number of words m in the original completion that the model is asked to distinguish from 1 to 6, since the difficulty may vary depending on the length of the original completion. We then compute the hypothetical consistency rate by calculating the proportion of the time that the model generated the letter of the answer choice corresponding to its own completion.

3.1 Experimental Setup

In all experiments, we evaluate four model sizes of the OpenAI GPT-3 model (Brown et al., 2020) – `ada-001`, `babbage-001`, `curie-001`, and `davinci-003` (in order of increasing capacity).[2] All experiments are run using greedily decoded completions obtained from the OpenAI API from Aug. 2022 to Jun. 2023. We use 0-shot initial prompts but evaluate hypothetical consistency prompts using k-shot prompts, where k ranges from 1 to 10. Since the in-context performance of LLMs is known to vary depending on the selection of in-context examples (Liu et al., 2021; Rubin et al., 2022), we randomly select a different set of in-context examples for each prompt. We also randomize the order of answer choices for each multiple-choice question to mitigate sensitivity to answer choice order (Pezeshkpour & Hruschka, 2023). Further evaluations for other combinations of models can be found in Appendix A.

Table 2: Average percent edit distances between completions from `davinci-003` versus completions from the three other models and two datasets.

Dataset	`davinci-003` / `ada-001`	`davinci-003` / `babbage-001`	`davinci-003` / `curie-001`	`davinci-003` / Dataset
Wikipedia	72.8%	69.7%	65.0%	70.3%
DailyDialog	75.8%	75.1%	74.2%	79.0%

3.2 All Model Sizes Perform Poorly At Distinguishing Their Own Completions

Figure 2 shows GPT-3's hypothetical consistency rates averaged over all few-shot prompts. Notably, all model sizes smaller than `davinci-003` perform at about random chance on this task, regardless of how many words of the original completion the model is tasked with predicting. `davinci-003` is the only model size that consistently performs above random chance, but even then its accuracy ranges only from 26% to 31% for Wikipedia and 30% to 37% for DailyDialog. We might also expect hypothetical consistency rate to increase as a function of m (because longer sequences provide more information for the model to use to distinguish its own completion from other sequences), but we do not observe this trend for these models. In Appendix A, however, we do observe this trend for `gpt4`. It appears that higher-capacity and generally more powerful models are overall more hypothetically consistent.

We also inspect the frequency with which each model selects each possible answer choice, as shown in Figure 3. For a highly hypothetically consistent model that has not memorized the dataset, we would expect it to

[2]We select this particular set of models since it is the most recent set of text completion (rather than chat) models available for each size of GPT-3. However, we also compare against `text-davinci-001` and `gpt4` in Appendix A. We separate `davinci-001` and `gpt4` into separate analyses because `davinci-001`, `davinci-003`, and `gpt4` often output the same original completions. In our multiple-choice set-up, this results in duplicate answer choices. As such, we only ever include completions from **one** of `davinci-001`, `davinci-003`, and `gpt4` in any hypothetical consistency prompt. Since `gpt4` is a chat model whereas the rest of the models are completion models, we choose to analyze the completion models in the main text and leave comparison against `gpt4` for the Appendix.